

EN401 — Handout 2: Energy Chain Data Structures

Activity Ratios, Conversion Efficiencies, and Grid Delivery Losses

1. Plant Conversion Efficiencies

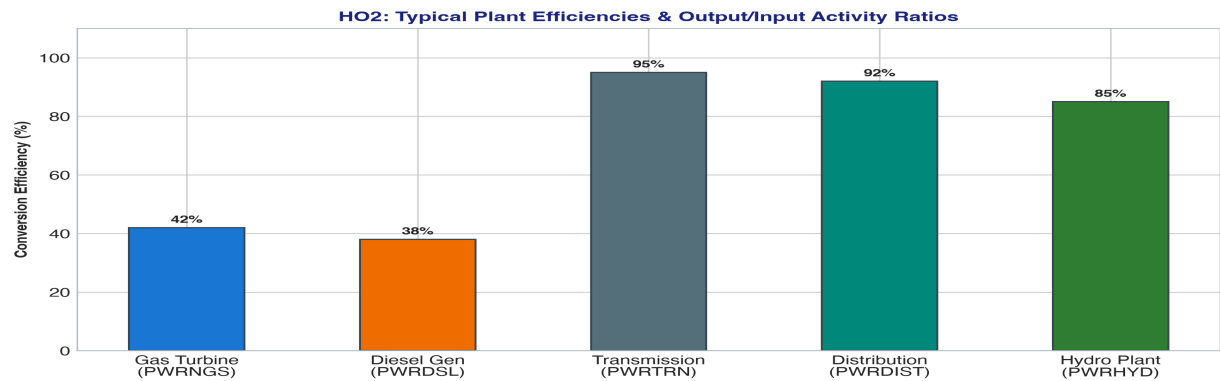


Figure: Conversion efficiencies across fossil and clean technologies.

2. Quantitative Activity Ratio Parameters

Tech Code	Fuel In	Input Ratio	Fuel Out	Output Ratio	Efficiency	Loss (%)
PWRNGS	NGS	2.857	ELC001	1.000	35.0%	65.0% Thermal Waste
PWRDSL	DSL	2.500	ELC001	1.000	40.0%	60.0% Thermal Waste
PWRHYD	HYD	1.000	ELC001	1.000	100.0%	0.0% Clean Conversion
PWRBIO	BIO	3.333	ELC001	1.000	30.0%	70.0% Thermal Waste
PWRTRN	ELC001	1.053	ELC002	1.000	95.0%	5.0% High-Voltage Loss
PWRDIST	ELC002	1.111	ELC003	1.000	90.0%	10.0% Low-Voltage Loss

3. Core Takeaways

- Defined InputActivityRatio to model thermal generation losses (35% gas, 40% diesel).
- Network losses accumulate: 95% transmission × 90% distribution = 85.5% cumulative grid efficiency.